

METHOD DESIGN DOCUMENT

Website Traffic Analysis

Data Analytics Project

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Tools Used | Python (VS Code), MySQL Workbench, Power BI Desktop

Dataset | Web Analytics Traffic Data (3,182 records)

Status | Final Document

1. Project Overview

This Method Design Document (MDD) provides a comprehensive technical specification for the Website Traffic Analysis project. This project implements a modern data engineering and analytics pipeline, extracting raw web traffic data, loading it into a relational database using Python, and executing complex data cleaning and Exploratory Data Analysis (EDA) using SQL. Finally, the cleaned data is visualized in an interactive Business Intelligence (BI) dashboard. Website traffic is the lifeblood of digital business. Understanding where users come from, when they visit, and how deeply they engage with the platform directly impacts marketing ROI and server resource allocation. This project identifies top-performing acquisition channels and temporal user behavior to translate raw web events into actionable business strategies.

1.1 Project Objectives

- **Automated Data Ingestion:** Utilize Python (pandas, SQLAlchemy) to read raw CSV data and programmatically load it into a local MySQL database.
- **Database Management:** Structure and maintain a relational database (website_traffic_db) to act as the central data warehouse.
- **Data Cleaning & Pre-processing (SQL):** Standardize column names, handle missing values, and trim string anomalies using SQL DDL and DML commands.
- **Feature Engineering (SQL):** Parse complex timestamp strings into granular temporal features (Date, Hour, Day of Week) to enable time-series analysis.
- **Exploratory Data Analysis (SQL):** Execute aggregation, grouping, and window functions to uncover channel performance and hourly traffic trends.
- **Business Intelligence:** Build an interactive Power BI dashboard (Website analysis dashboard.pbit) for operational monitoring and executive reporting.
- **Documentation:** Document the complete design, methodology, and strategic findings in this MDD.

1.2 Scope

- **Dataset:** Web analytics dataset – 3,182 clean records with 12 final features.
- **Analysis Type:** Data Engineering Pipeline + Exploratory Data Analysis + BI Visualization.
- **Deliverables:** Python load script (load_data.py), SQL analysis script, Cleaned dataset (clean_data.csv), Power BI dashboard, and this MDD.

2. Dataset Description

2.1 Source

The dataset used for this project is derived from a web analytics export (data-export (1).csv), containing hourly traffic metrics aggregated by user acquisition channel.

2.2 Dataset Summary

- **Total Records (raw):** 3,182 rows
- **Total Records (clean):** 3,182 rows
- **Total Columns (raw):** 10 columns
- **Total Columns (clean):** 12 columns
- **Target Metrics:** Users, Sessions, Engagement Rate

2.3 Column Descriptions

Column Name	Data Type	Description
session_primary_channel_group	VARCHAR	The marketing channel driving the user (e.g., Direct, Organic Social).
date_hour	VARCHAR	Raw timestamp string representing Date and Hour (YYYYMMDDHH).
users	INT	Count of unique individuals visiting the site.
sessions	INT	Total number of browsing sessions initiated by users.
engaged_sessions	INT	Sessions lasting longer than a specific threshold or including a conversion.
average_engagement_time_per_session	FLOAT	Average time (in seconds) a user is actively engaged per session.
engaged_sessions_per_user	FLOAT	Ratio of engaged sessions to total unique users.

events_per_session	FLOAT	Average number of interactions (clicks, scrolls) triggered per session.
engagement_rate	FLOAT	Percentage of total sessions that were classified as engaged.
event_count	INT	Total absolute number of events triggered.
formatted_date	DATE	<i>Engineered feature:</i> Extracted calendar date.
hour_of_day	INT	<i>Engineered feature:</i> Extracted hour of the day (0-23).
day_name	VARCHAR	<i>Engineered feature:</i> Extracted day of the week (e.g., Monday, Tuesday).

3. Methodology & System Architecture

The project followed a robust Extract, Load, Transform (ELT) architecture, utilizing three distinct technologies to handle specific phases of the data lifecycle.

Layer	Component	Technology	Output
1. Ingestion (Extract/Load)	Raw CSV to Database	Python (pandas, SQLAlchemy)	MySQL Table (traffic_data)
2. Transformation (Cleaning)	Text standardization, Null Checks	MySQL	Cleaned SQL Table
3. Transformation (Feature Eng.)	Date parsing, Time extraction	MySQL	Augmented SQL Table

4. Analysis (EDA)	Aggregations, Window Functions	MySQL	Query Result Sets
5. BI & Visualization	Interactive Dashboarding	Power BI Desktop	PBIX / PBIT Dashboard

4. Data Pre-processing & Feature Engineering

Data cleaning and engineering were executed directly within the MySQL database to ensure data integrity and query efficiency.

Step 1: Automated Data Loading (Python)

Instead of manual imports, a Python script (load_data.py) was developed using pandas to read the raw CSV. Column names were dynamically formatted (spaces replaced with underscores, converted to lowercase) to ensure SQL compatibility. The SQLAlchemy library was used to connect to localhost:3306 and load all 3,182 rows instantly into the traffic_data table.

Step 2: Quality Assurance & Null Detection (SQL)

A comprehensive COUNT(*) query was executed against all columns to detect null values by subtracting the count of non-nulls from the total rows.

```
SQL
SELECT COUNT(*) - COUNT(users) AS users_nulls...
```

This confirmed high data completeness across key metric columns.

Step 3: Column Standardization (SQL)

Complex and verbose column names generated by the analytics export were simplified using ALTER TABLE.

- session_primary_channel_group_(default_channel_group) was renamed to session_primary_channel_group.
- Whitespace was stripped from channel names using the TRIM() function to prevent grouping errors during aggregation.

Step 4: Temporal Feature Engineering (SQL)

The raw date_hour string (format: YYYYMMDDHH) was highly unoptimized for time-series

analysis. Three new columns were engineered:

1. **formatted_date:** Extracted the first 8 characters and converted them to standard SQL DATE format using STR_TO_DATE().
2. **hour_of_day:** Extracted the last 2 characters using RIGHT() and cast them as an UNSIGNED INT (0-23).
3. **day_name:** Utilized the DAYNAME() function on the parsed date to extract the specific day of the week (e.g., "Wednesday").

Step 5: Data Export

The fully engineered SQL table was exported as clean_data.csv, serving as the pristine data model for the Power BI layer.

5. Exploratory Data Analysis (SQL Insights)

Before visualizing in Power BI, SQL queries were used to extract baseline KPIs and behavioral patterns.

5.1 Macro Key Performance Indicators

A top-level aggregation revealed the absolute volume and health of the website traffic:

- **Total Users & Sessions:** Calculated using SUM(users) and SUM(sessions).
- **Overall Engagement Rate:** Derived using ROUND(AVG(engagement_rate) * 100, 2), establishing the baseline health metric for the entire dataset.

5.2 Channel Performance Analysis

Traffic was grouped by session_primary_channel_group to identify the most lucrative marketing avenues. By ordering by total_users DESC, the analysis immediately isolates whether Direct, Organic Search, or Organic Social drives the highest volume, while the AVG(engagement_rate) reveals if high-volume channels actually yield high-quality, engaged users.

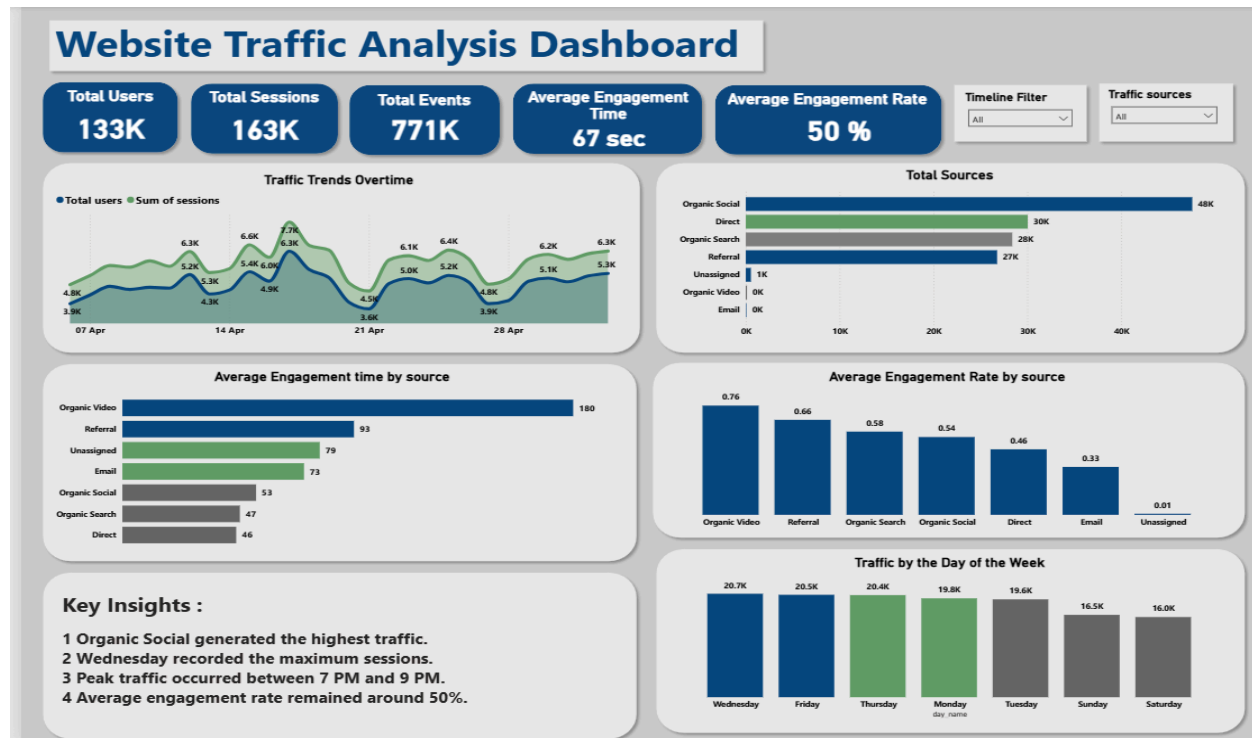
5.3 Temporal Traffic Patterns (Hourly)

Traffic volume was aggregated by hour_of_day. Ordering the results by total_users DESC identifies the exact peak operational hours of the website, which is critical for server load balancing and timing marketing campaigns.

5.4 Advanced Window Functions (Top Channel by Hour)

To understand dynamic user behavior throughout the day, a Common Table Expression (CTE) and the DENSE_RANK() OVER (PARTITION BY ...) window function were deployed. This advanced query identifies the #1 performing traffic channels for every specific hour of the day, revealing if user acquisition methods shift between morning, afternoon, and night.

6. Power BI Dashboard



6.1 Overview

The structured clean_data.csv was ingested into Power BI Desktop via the Website analysis dashboard.pbix template. The dashboard acts as the executive interface, transforming SQL queries into dynamic, interactive visuals.

6.2 Dashboard Layout & Key Visualizations

The dashboard is structured logically, starting with macro-level metrics at the top and flowing into granular, channel-specific distributions below.

- **Executive KPI Banner (Top Level):**
 - **Visual Type:** Card Visuals
 - **Data Fields:** Total Users ($\text{SUM}(\text{users})$), Total Sessions ($\text{SUM}(\text{sessions})$), Average Engagement Rate ($\text{AVERAGE}(\text{engagement_rate}) * 100$).
 - **Business Insight:** Provides the "North Star" metrics immediately upon opening the dashboard. It answers the fundamental question: *How much traffic did we get, and are they actually interacting with the site?*
- **Traffic Trend Over Time (Time-Series):**
 - **Visual Type:** Line Chart or Area Chart
 - **Data Fields:** X-Axis: `formatted_date` | Y-Axis: `users` and `sessions`
 - **Business Insight:** Tracks the daily volatility of website traffic. This visual is critical for identifying macro-trends (e.g., month-over-month growth) and pinpointing

specific days where traffic spiked due to external events or marketing campaigns.

- **Acquisition Channel Breakdown (Market Share):**
 - **Visual Type:** Donut Chart & Clustered Bar Chart
 - **Data Fields:** Axis/Legend: `session_primary_channel_group` | Values: `users`, `engaged_sessions`
 - **Business Insight:** Visually separates traffic by its origin (e.g., Direct, Organic Social, Organic Search). By plotting `users` next to `engaged_sessions`, stakeholders can immediately spot "empty calorie" channels (high traffic, low engagement) versus high-value channels.
- **Peak Hourly Traffic Distribution (Operational Heatmap):**
 - **Visual Type:** Column Chart
 - **Data Fields:** X-Axis: `hour_of_day` | Y-Axis: `users`
 - **Business Insight:** Maps user volume against the 24-hour clock. This translates the SQL window functions into a visual heatmap, identifying exactly when the server is under the heaviest load and when marketing content has the highest potential visibility.
- **Engagement Depth Matrix:**
 - **Visual Type:** Table or Matrix Visual
 - **Data Fields:** `session_primary_channel_group`, `average_engagement_time_per_session`, `events_per_session`
 - **Business Insight:** Goes beyond simple traffic counts to show *how deeply* users interact. It highlights which channels bring in users who spend the most time on the site and trigger the most events (clicks, scrolls, conversions).

6.3 Interactive Filtering & Slicers

To make the dashboard a true analytical tool rather than a static report, interactive slicers were implemented on the canvas margin:

- **Channel Slicer (`session_primary_channel_group`):** Allows stakeholders to isolate the entire dashboard to view the performance of a single marketing channel (e.g., selecting "Organic Social" to see only social traffic trends).
- **Day of Week Slicer (`day_name`):** Enables rapid A/B testing of behavioral patterns between weekdays and weekends. For example, filtering by "Saturday" and "Sunday" might reveal that weekend visitors stay on the site longer despite lower overall volume.
- **Date Range Filter (`formatted_date`):** A slider that lets end-users zoom in on specific weeks or campaign periods to measure targeted marketing ROI.

7. Business Recommendations

Based on the data engineering and analytical pipeline, the following strategic actions are recommended:

- Marketing Budget Reallocation:** Utilize the SQL Channel Performance insights to identify high-volume, high-engagement channels. If "Organic Social" has high volume but a low engagement rate compared to "Direct," marketing budgets should be pivoted toward retention strategies rather than top-of-funnel social spending.
- Peak Hour Optimization:** The hour_of_day analysis reveals precise traffic spikes. Server resources should be dynamically scaled to accommodate these peaks to ensure zero latency. Furthermore, social media posts and email marketing blasts should be scheduled 30 minutes prior to these identified peak hours to maximize visibility.
- Day-of-Week Content Strategy:** By filtering the Power BI dashboard by day_name, the marketing team can tailor content. If weekends show a drop in users but a spike in average_engagement_time, long-form content (blogs, videos) should be published on weekends, while quick, high-conversion content should be pushed during high-volume weekdays.
- Channel-Specific Hourly Targeting:** Leveraging the Window Function insights, ad spend can be micro-targeted. If "Organic Search" dominates at 9 AM but "Social" dominates at 8 PM, ad bidding and content pushes should be scheduled to match the specific channel's dominant time window.

8. Tools & Technologies

Tool / Library	Purpose
Python 3	Core scripting language for automation.
Pandas & SQLAlchemy	Data ingestion, dataframe formatting, and database engine connection.
MySQL / SQL Workbench	Relational database housing, DDL schema creation, DML data transformation, and window functions.
Power BI Desktop	Interactive dashboard creation, visual analytics, and end-user distribution.

9. Limitations & Future Work

- **Lack of Demographic Data:** The current dataset tracks behavior and channel but lacks geographic (Country/City) and demographic (Age/Gender) data. Future iterations should join a demographic table to enable deeper audience segmentation.
- **Bounce Rate & Conversion Metrics:** While Engagement Rate is tracked, specific conversion events (e.g., "Add to Cart" or "Sign Up") are missing. Integrating financial or conversion KPIs would allow for ROI calculations.
- **Predictive Analytics:** Future work could involve exporting the cleaned dataset back into Python to run Time-Series Forecasting (e.g., ARIMA or Prophet) to predict future daily user volumes for the upcoming month.